Shortening the Short Course of Tuberculosis Treatment

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One of the great satisfactions of managing infectious diseases is the remarkable and rapid efficacy of antibiotics. The first uses of penicillin in the treatment of pneumococcal pneumonia produced near miraculous results, and most bacterial infections can be cured with a few days of therapy. However, tuberculosis has been an outlier. A series of landmark studies performed over several decades showed that combination therapy could result in high rates of cure among patients with tuberculosis, but the best "short course" regimen — and the one that remains in use to this day — still requires 6 months of therapy with multiple drugs.[1]

Many efforts have been made to shorten this period. There is good evidence that this may be possible, because most patients who receive standard therapy are cured well before 6 months. The longer duration is driven by a minority of patients for whom extended therapy is warranted. A "stratified medicine" approach proposed recently would entail identifying those in need of longer treatment.[2] The alternative approach is to replace the standard regimen with one that provides a durable cure for all patients in less time. However, three large studies that evaluated various fluoroquinolone-containing regimens all showed that the tested 4-month regimens were not noninferior to the 6-month course of treatment.[3-5] It was almost as though 6 months represented some strict limit — that is, until now.

In this issue of the *Journal*, the elegant study by Dorman et al.[6] tested a prediction derived from preclinical studies in animals that higher doses of rifamycins could increase the rate of clearance of infecting bacteria in tuberculosis. Thus, they designed a three-group, open-label trial of rifapentine, an alternative rifamycin that has a longer half-life than rifampin, the standard drug. Participants were randomly assigned to the standard 6-month regimen (8 weeks of once-daily rifampin, isoniazid, pyrazinamide, and ethambutol, followed by 18 weeks of once-daily rifampin and isoniazid [the control group]) or one of two experimental 4-month regimens (8 weeks of once-daily rifapentine, isoniazid, pyrazinamide, and ethambutol, followed by 9 weeks of once-daily rifapentine and isoniazid [the rifapentine group], or the same regimen but with the fluoroquinolone moxifloxacin substituted for ethambutol, followed by 9 weeks of once-daily rifapentine, isoniazid, and moxifloxacin [the rifapentine–moxifloxacin group]). The primary efficacy outcome was survival free of tuberculosis at 12 months, and the responses to treatment were ranked according to three sets of criteria (favorable, unfavorable, and not assessable); survival free of tuberculosis at 18 months was a secondary outcome. The investigators also collected information on the safety of all the regimens. The trial was designed to test the noninferiority of the two 4-month regimens to the standard 6-month regimen with respect to the primary outcome; noninferiority was shown if the upper bound of the 95% confidence interval of the between-group difference was 6.6 percentage points or less. The investigators used a hierarchical analysis that allowed them to test the noninferiority of the rifapentine regimen if the rifapentine–moxifloxacin regimen was shown to be noninferior.

A total of 2343 participants were enrolled in the trial at 34 sites from around the world. In general, the incidences of adverse events were fairly similar in the three treatment groups. The rifapentine–moxifloxacin group met the criteria for noninferiority. In one of two primary analysis populations, an unfavorable outcome occurred in 15.5% of the participants in the rifapentine–moxifloxacin group and in 14.6% in the control group (adjusted difference, 1.0 percentage points; 95% confidence interval [CI], –2.6 to 4.5), and in the other primary analysis population, unfavorable outcomes occurred in 11.6% and 9.6%, respectively (adjusted difference, 2.0 percentage points; 95% CI, –1.1 to 5.1). However, although close, noninferiority was not shown for the rifapentine regimen. Thus, a trial has shown that 4 months of treatment can be similar to standard therapy for 6 months.

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What does this mean for the future of tuberculosis therapy? There are both immediate and longer-term implications. Both rifapentine and moxifloxacin are widely available and could probably be packaged appropriately for use by national tuberculosis programs. Shortening a regimen by 2 months would make treatment somewhat less cumbersome and probably make it more cost-effective. However, the infrastructure required to ensure adherence would be largely unchanged. The need to take rifapentine after meals to maximize absorption could introduce new issues with adherence. Moreover, one of the advantages of the currently used tuberculosis drugs is that they are not widely used in other infections. In addition to necessitating rapid drug-susceptibility testing for moxifloxacin, widespread use of this antibiotic for the treatment of tuberculosis could promote resistance to fluoroquinolones in other bacteria.

This trial does, however, establish an important principle: there is no magic with 6 months of therapy. We do not know what the biologic limits of therapy are, but it might be possible to get to shorter regimens. The inability of an early biomarker, time to sputum culture conversion, to predict outcomes is disappointing. However, although the development of the standard 6-month regimen resulted from trials in humans, the rifapentine–moxifloxacin therapy was first shown to be effective in shortening therapy in animals, a finding that suggests that the development of new therapies is on an accelerated path. Indeed, additional new regimens have proved to be far better in mice than the one tested in this trial.[7] Thus, this trial not only proves that we can have a shorter short-course treatment but also suggests that an even shorter short-course treatment might one day be feasible.

Disclosure forms provided by the authors are available with the full text of this editorial at NEJM.org.

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